

Boolean functions

$$f: \{0,1\} \mapsto \{0,1\}$$

or in general

$$f: \{0,1\}^n \mapsto \{0,1\}$$

1 bit function:

e.g. $f(0) = 0$

$$f(0) = 0$$

$$f(1) = 0$$

constant

$$f(0) = 0$$

$$f(1) = 1$$

$$f(0) = 1$$

$$f(1) = 0$$

Balanced

In the quantum setting:

f as a unitary op:

$$f|0\rangle = |0\rangle$$

e.g.

$$f|1\rangle = |0\rangle$$

Problem:

Superposition:

Consider Constant
Zero Function

$$f\left(\frac{1}{\sqrt{2}}|0\rangle + |1\rangle\right) =$$

$$= \frac{1}{\sqrt{2}} (f|0\rangle + f|1\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |0\rangle)$$

$$= \frac{2}{\sqrt{2}} |0\rangle \rightarrow \text{Not Valid}$$

Solution: enlarge the Hilbert space

$$\mathcal{F}(|+\rangle \otimes |0\rangle)$$

$$= \frac{1}{\sqrt{2}} (\mathcal{F}|0\rangle|0\rangle + \mathcal{F}|1\rangle|0\rangle)$$

$$= \frac{1}{\sqrt{2}} |0\rangle|f(0)\rangle + \mathcal{F}|1\rangle|f(1)\rangle$$

Generalization:

$$\mathcal{F} |a\rangle |b\rangle = |a\rangle |b \oplus f(a)\rangle$$

Addition modulo 2

What quantum gate could work as a function?

CNOT as a function

$$\text{CNOT } |a\rangle |b\rangle = |a\rangle |b \oplus a\rangle$$

$$|a\rangle |b\rangle \mapsto |a\rangle |b \oplus a\rangle$$

$$|0\rangle |0\rangle \mapsto |0\rangle |0\rangle$$

$$|0\rangle |1\rangle \mapsto |0\rangle |1\rangle$$

$$|1\rangle |0\rangle \mapsto |1\rangle |1\rangle$$

$$|1\rangle |1\rangle \mapsto |1\rangle |0\rangle$$

CNOT acts
as a
XOR

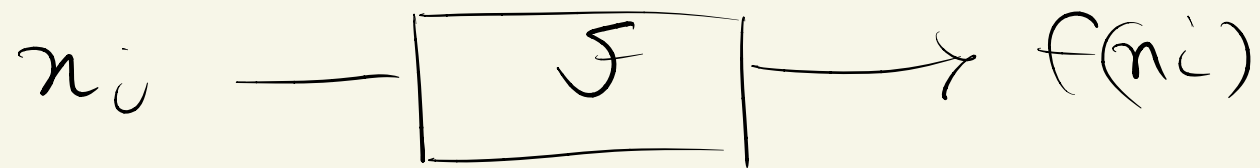
$$\text{CNOT} = |+\rangle |0\rangle$$

$$= \frac{1}{\sqrt{2}} |00\rangle + |11\rangle$$

CNOT acts as a balanced function.

Now, imagine you have a function f
as a black box.

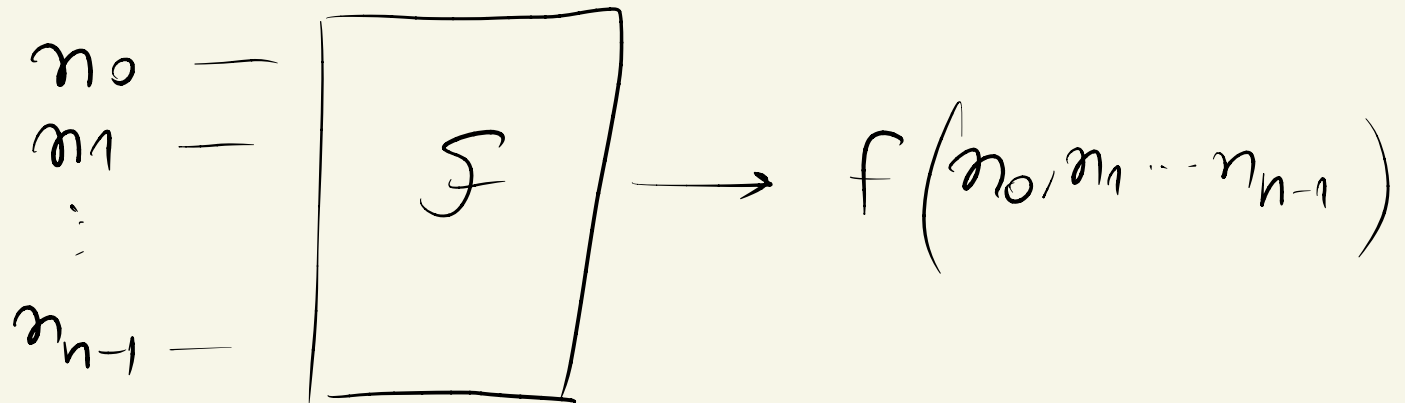
$$f: 20,19 \mapsto 20,14$$



How many calls to the function do
you need to clarify it as 'constant/balanced'?

What if

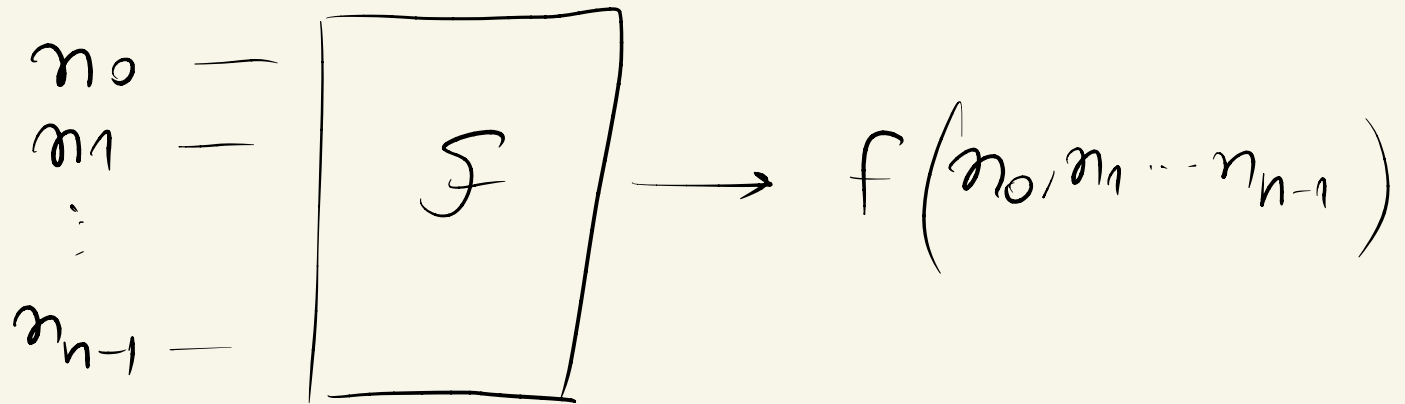
$$f: \{0,1\}^n \mapsto \{0,1\}$$



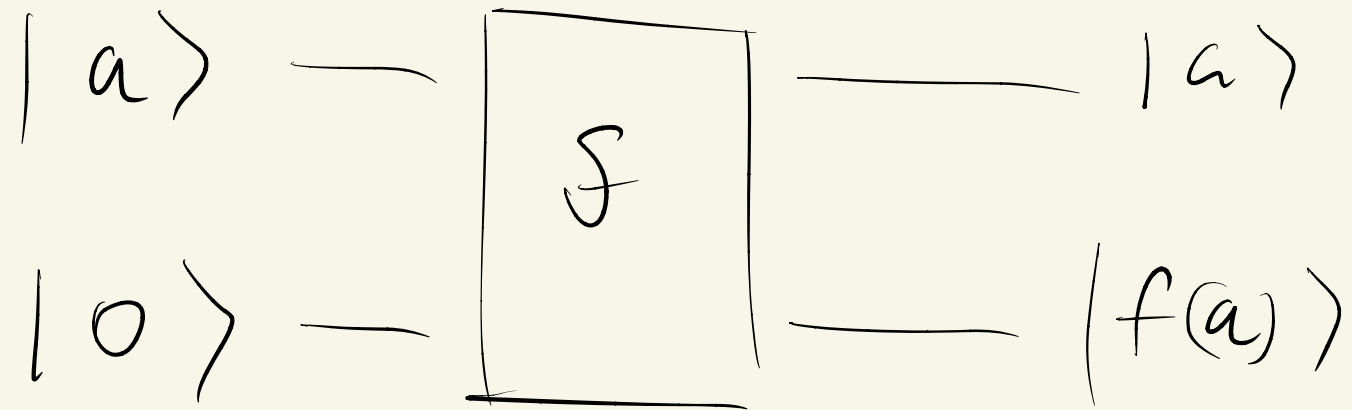
How many cells?

What if

$$f: \{0,1\}^n \mapsto \{0,1\}$$



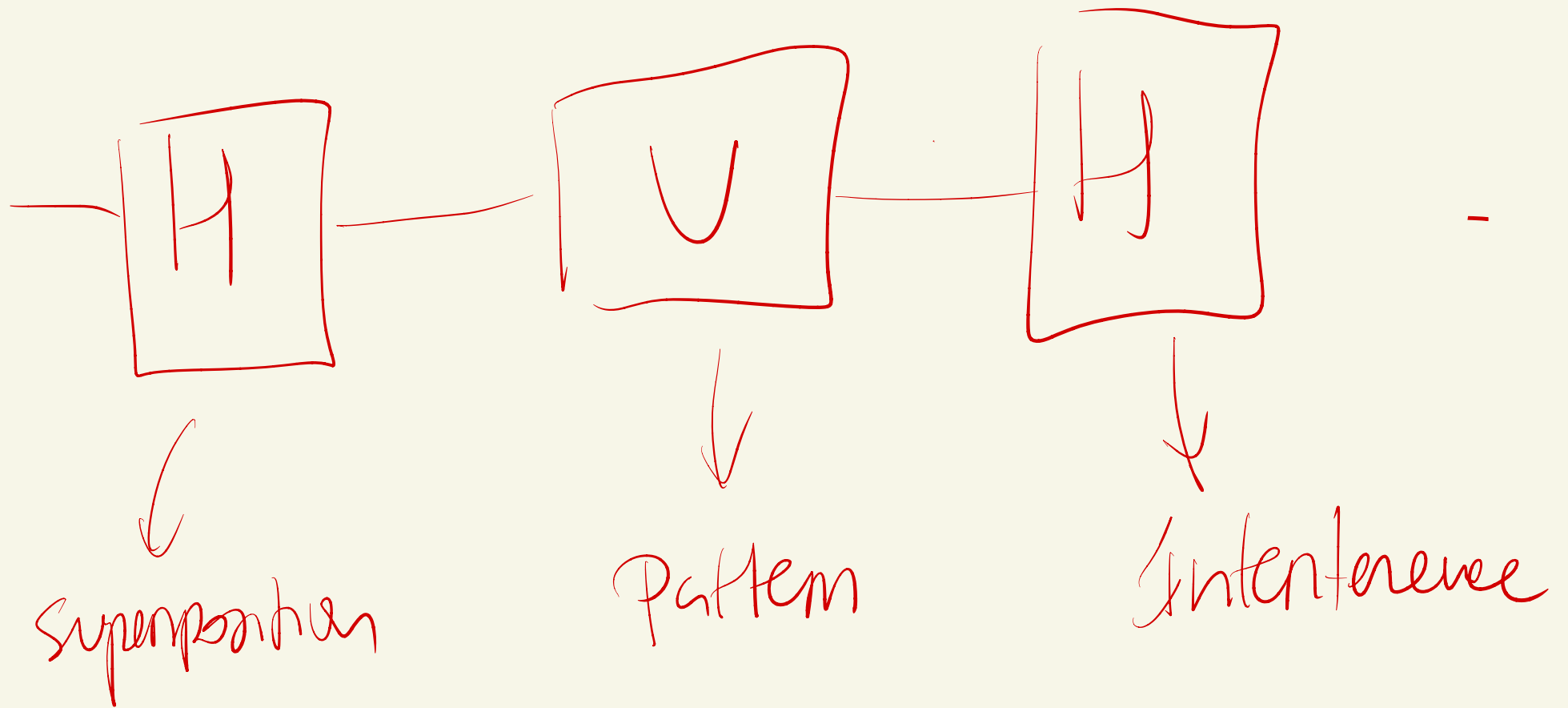
How many cells? $O(2^n)$



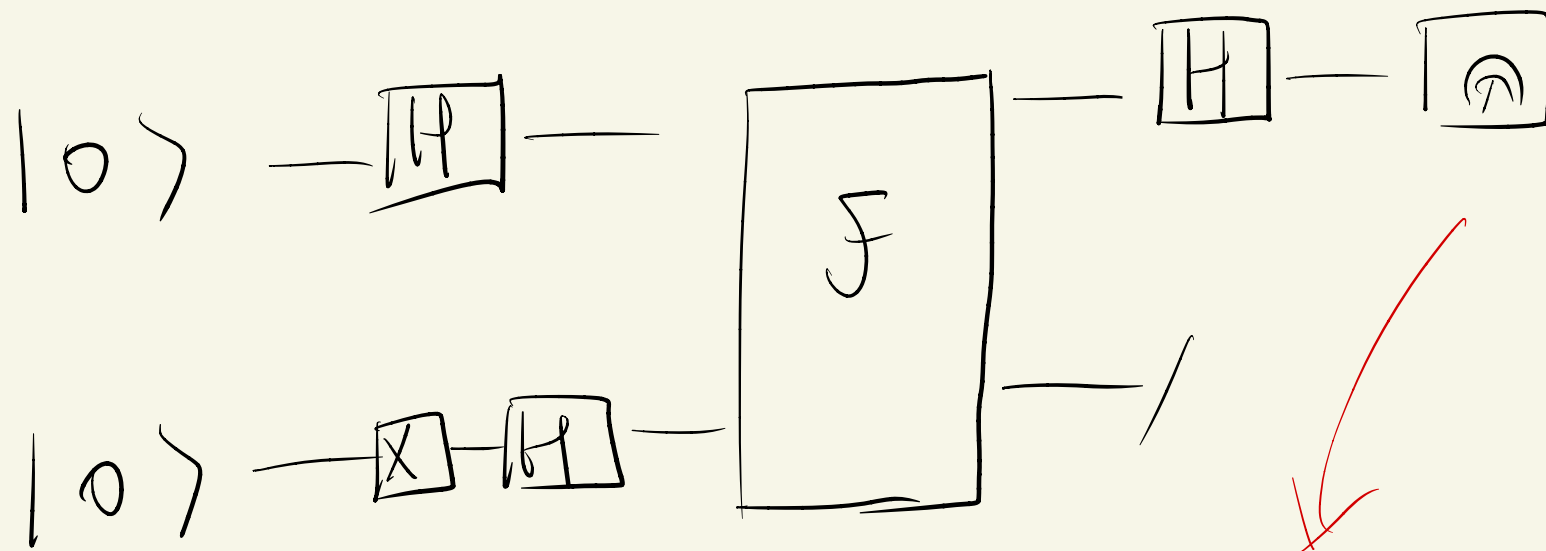
Quantum computing can do it
with a single cell!

How?

Recall the golden pattern

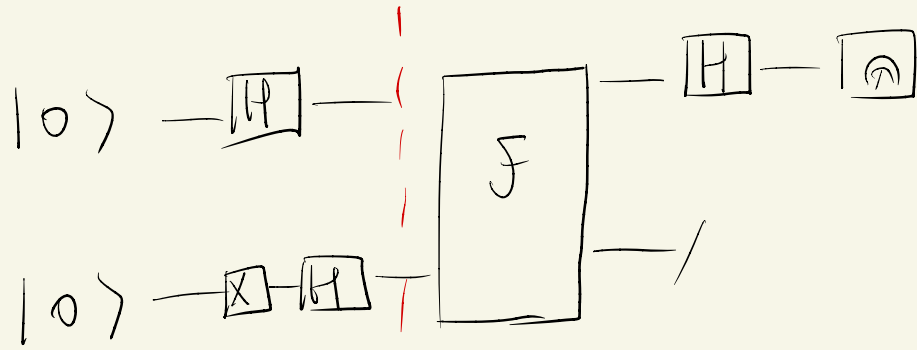


Deutsch Algorithm



if $|0\rangle = \text{constant}$
if $|1\rangle = \text{balanced}$

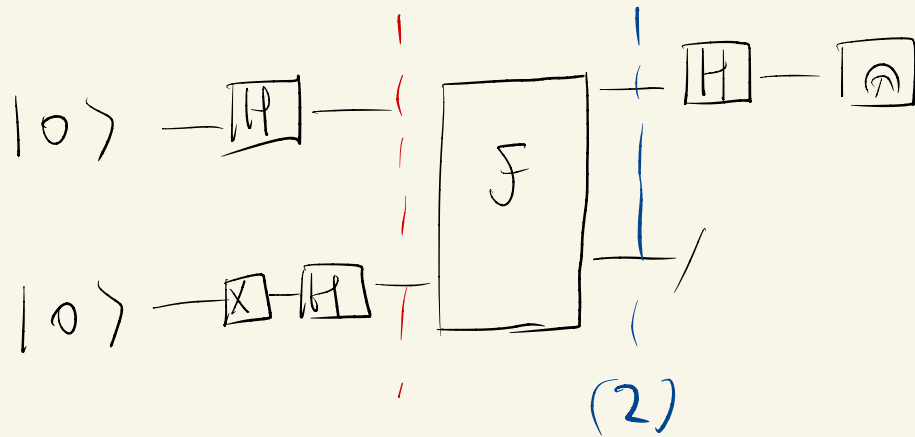
$\mathcal{F} = \text{CNOT} \rightarrow \text{balanced} \rightarrow \text{output } |1\rangle$



$$(1) \quad = |+\rangle |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= \frac{1}{\sqrt{2}}|0\rangle \left(\frac{1}{\sqrt{2}}|0\rangle - |1\rangle \right) + \frac{1}{\sqrt{2}}|1\rangle \left(\frac{1}{\sqrt{2}}|0\rangle - |1\rangle \right)$$

$f = \text{CNOT} \rightarrow \text{balanced} \rightarrow \text{output } 11$



$$(1) = \frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) + \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right)$$

$$(2) = \text{CNOT} \left(\frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) + \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) \right)$$

$$\frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) + \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |1\rangle - |0\rangle \right)$$

$$\frac{1}{\sqrt{2}} (|1\rangle - |0\rangle) = -\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Phase Kickback

$$\frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) - \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right)$$

$$\frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) + \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |1\rangle - |0\rangle \right)$$

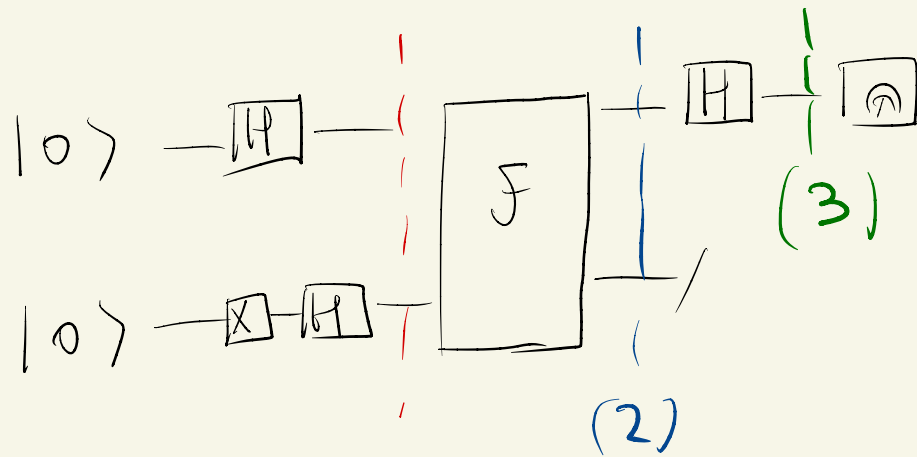
$$\frac{1}{\sqrt{2}} (|1\rangle - |0\rangle) = -\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Phase Kickback

$$\frac{1}{\sqrt{2}} |0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right) - \frac{1}{\sqrt{2}} |1\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - |1\rangle \right)$$

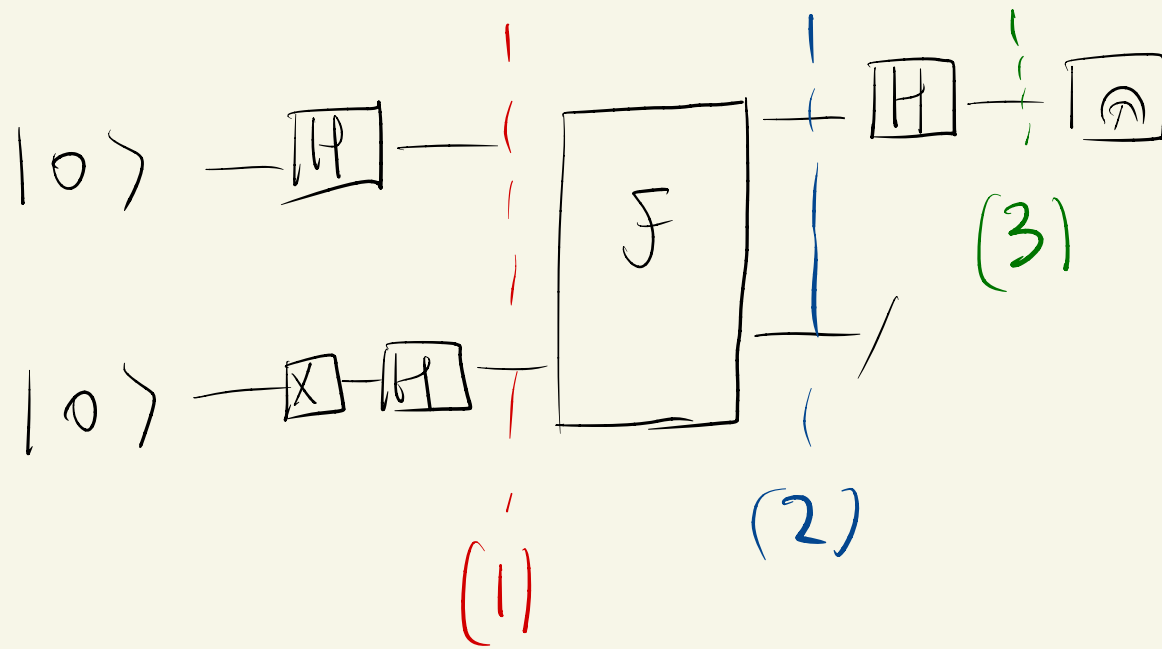
$\underbrace{\hspace{10em}}_{|-\rangle}$
 $\underbrace{\hspace{10em}}_{|-\rangle}$

$$= |-\rangle |-\rangle$$



$$(3) \quad (H \otimes I) |-\rangle |-\rangle = |1\rangle |-\rangle$$

Deutsch Algorithm



$|0\rangle$ then constant
 $|1\rangle$ then balanced

(1) Superposition

(2) Ancilla $\rightarrow$
 Phase Kickback

(3) Interference